

Math5485

Introduction to Numerical Methods I

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Welcome!

These are the lecture notes for Math 5485: Introduction to Numerical Methods I, Fall 2026.

Use the links on the side or navigate by chapter:

- [Chapter 1: An Introduction to Numerical Analysis](#)
- [Chapter 2: Solving Nonlinear Equations in 1d](#)
- [Chapter 3: Polynomial Interpolation](#)
- [Chapter 4: Numerical Integration](#)
- [Chapter 5: Linear systems of equations](#)

Useful links:

[Syllabus](#) [GitHub](#) [Fall 2025 Course](#) [Ask a Question](#) [Suggest Correction](#) [Math 5486](#)

References

- [Driscoll and Braun \(2017\)](#)
- [Burden et al. \(2016\)](#)
- [Süli and Mayers \(2003\)](#)
- [Trefethen \(2019\)](#)

Part I

Lectures

Part II

Problem Classes

Part III

Assignments

Part IV

Exams

Resources

Burden, Richard L., J. Douglas Faires, and Annette M. Burden. 2016. *Numerical Analysis*. 10th ed. Cengage Learning. <https://doi.org/10.13140/2.1.4830.2406>.

Driscoll, Tobin A., and Richard J. Braun. 2017. *Fundamentals of Numerical Computation*. Society for Industrial; Applied Mathematics. <https://doi.org/10.1137/1.9781611975086>.

Süli, Endre, and David F. Mayers. 2003. *An Introduction to Numerical Analysis*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511801181>.

Trefethen, Lloyd N. 2019. *Approximation Theory and Approximation Practice, Extended Edition*. Society for Industrial; Applied Mathematics. <https://doi.org/10.1137/1.9781611975949>.